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Talk

“TMR SENSORS Offer Better Signal-To-Noise Ratio And Faster Response THAN HALL SENSORS”

With unmatched precision and speed, how are TMR sensors set to transform current sensing and drive the future of automation and EVs? Ram Santhappan of Allegro MicroSystems explains their innovations to EFY's Nidhi Agarwal.

How are TMR sensors different from traditional current sensors?

Our sensors are based on XtremeSense Tunnelling Magnetoresistance (TMR) technology, which identifies the magnetic fields generated by current flowing in a circuit. By avoiding the energy loss of traditional sensors that use shunt resistors to measure current, TMR sensors offer faster response times, higher efficiency, and better accuracy by detecting minute magnetic-field changes. They also exhibit minimal hysteresis, ensuring consistent, repeatable measurements.

How are they better than Hall sensors?

You see, the TMR sensors have a better signal-to-noise ratio (SNR), making them more precise and sensitive for detection. The technology also responds faster, with less delay, which is crucial for protecting expensive electronics using gallium nitride (GaN) and silicon carbide (SiC) field-effect transistors (FETs) that require rapid response to current changes. While Hall sensors can support up to 5MHz bandwidth, TMR sensors can operate effectively at higher frequencies.

How are they suitable for GaN and silicon carbide devices?

TMR sensors are essential in safeguarding delicate components of GaN and SiC devices with better SNR, as wide band gap switches demand rapid current measurements to avoid catastrophic damage. If the circuit draws too much current and there is a delay in response, the switches can be permanently damaged. Enter TMR sensors—high-speed, precision devices that quickly detect and respond to current fluctuations, ensuring switches remain safe and operational.

What are the potential applications of TMR sensors?

Our new sensor measures up to 1000 amps of current directly on a circuit board without the need for a magnetic core. Compact and sleek, it fits effortlessly onto the board, enabling applications across power supplies, medical devices, aerospace, and industrial systems. The main users include data centres that are essential to artificial intelligence (AI). By swiftly detecting overcurrent situations it can shut down equipment before any damage occurs, protecting valuable machinery, whether in motors, robotics, or any high-current environments.

What hurdles do you encounter while designing?

To save space, we aim to make

the chips as small as possible. But every innovation has its own set of obstacles. One of the major challenges we face is managing the internal lead frame, as the current flows through it due to the integrated current sensor. To ensure the sensor is sensitive enough for accurate detection, extensive testing and simulations are required. Additionally, with millions of sensors, each must consistently meet a specific error range. The manufacturing process is complex and requires strict standards for materials.

What are the trade-offs in designing?

It depends on the product's environment. One can assess the noise level and available space to decide between a bus bar or routing on a circuit board. If space is tight, you can opt for a smaller design. It is the surface-mount packaging that makes our product better, eliminating the need for an external current sensor module.

How is making TMR sensors different from manufacturing Hall sensors?

As I mentioned, TMR sensors offer better SNR and faster response than Hall sensors but the manufacturing of TMR sensors is